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Started	Wednesday, 14 October 2026, 9:01 PM
Completed	Wednesday, 14 October 2026, 9:09 PM
Duration	7 mins 30 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 200,470,80,150,40,90,300,400,70,120

b. 40,70,80,90,120,150,200,300,400,470

c. 80,150,200,470,40,90,300,400,70,120

d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **2**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the largest element to its correct position

b. Bubbling up the smallest element

c. Sorting the entire list in one pass

d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **3**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Merge Sort
- b. Bubble Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes data harder to manage
- b. Decreases the efficiency of algorithms
- c. Provides a basis for other algorithms to work efficiently
- d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Backtracking
- b. Dynamic programming
- c. Divide and conquer
- d. Greedy method

The correct answer is:
Divide and conquer

Question **6**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. order()
- b. sorted()
- c. arrange()
- d. sort()

The correct answer is: sort()

Question **7**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Finding a specific element in a list
- b. Arranging data in a specific order
- c. Removing duplicates from a list
- d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **8**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Bubble sort
- b. Quick sort
- c. Insertion sort
- d. Heap sort

The correct answer is:
Insertion sort

Question **9**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are merged without sorting
b. Each part is sorted independently
c. The parts are ignored
d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **10**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data in a single pass
b. Sorting data sequentially
c. Rearranging data without sorting
d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **11**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It bubbles up the largest element in each pass
b. It uses the heap data structure
c. It divides the list into sublists
d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass

Question **12**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Searching

b. Rearranging

c. Sorting

d. Merging

The correct answer is: Sorting

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items

b. Sorting helps in finding duplicates quickly

c. Sorting decreases the efficiency of selection operations

d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **14**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It makes the analysis slower

b. It allows for quicker identification of duplicates

c. It complicates the analysis process

d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **15**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. All of the above
- b. Selection
- c. Duplicates
- d. Searching

The correct answer is: All of the above